

So what's this driver i hear about?

Till now, we've seen the full-range driver, generally used in radios and laptop speakers. Unfortunately, these systems are unable to reproduce the entire range of audible frequencies, and are useful only for space constrained applications, or paces where extremely faithful reproductions of sound are not required. These would be public announcement systems, TVs etc. To overcome their inability to reproduce extremely high frequency sounds, full-range drivers often employ an additional cone called a whizzer: a small, light cone attached to the joint between the voice coil and the primary cone. The whizzer cone effectively increases the response of the driver for higher frequencies, broadening its high frequency directivity.

A single hi-fidelity speaker system will contain three to four types of drivers – a tweeter, that reproduces treble frequencies and sounds on the higher range of the spectrum; woofers, that reproduce the bass and low ranges; and midrange drivers that specialize in producing midrange frequencies.

The woofer produces bass sounds, typically from 60 Hz up to 250 Hz, while the tweeter, or treble speaker, will produce high audio frequencies, typically from around 2,000 Hz to 20,000 Hz. The mid-range speaker reproduces sound in the frequency range from 250 Hz to 2000 Hz. These three form an entire speaker system, and are generally combined into one single complete loudspeaker enclosure to provide clean and high quality performance. Additionally, some systems divide the range of frequencies between the woofer and tweeter itself, by reducing the lower limit of the tweeter's sound and increasing the upper limit of the woofer, thus eliminating the mid-range driver. This is done to introduce the sub-woofer, another special driver that focuses on reproducing EXTREMELY low frequencies, many of which are not audible for humans. This driver's purpose is to add more shake and roll to the listening experience, which is done in concerts and home theatre systems, where, at louder volumes, you can feel the bass and impact of the instruments. The main purpose of subwoofers is to recreate this experience without the need for the entire playback to be at high volume.

Another component, used mainly in recording studios, filmmaking, television studios, radio studios, is a monitor speaker small enough to sit on a stand or desk closer to the listener. This lets the listener hear the sound emanating directly from the speaker, rather than being reflected from other surfaces in the room. These studio monitors are able to give an accurate reproduction of the tonal qualities of the audio, without any distortion in the sound-stage perspective for stereo recordings.